LOOMSHIELD -- Brand Governance Platform

Project Submission Specification & Verification Report

Project Name: LoomShield	**Team Name:** isaac.incognito007 (Solo Team)	**Participant:** Isaac Shiloah Herbert V
Source-Code Repo: https://github.com/isaacincognito007-cloud/Inkloom-artLoomShield	**Email:** isaac.incognito007@gmail.com	**Working project link:** https://inkloom.art
Operating Mode: Two Local On-Device Engines (Deterministic + NLP)	**API Key Policy:** Zero Mandatory External API Keys (100% On-Device)	**Automated Tests:** 68/68 automated tests passed (100% pass rate); code coverage measured separately.

Executive Summary: LoomShield is an enterprise-grade, local-first brand governance workspace that inspects multi-format marketing content against approved brand guidelines before publication. Built with two on-device engines (Tier 1 Deterministic + Tier 2 Embedded Scikit-Learn TF-IDF NLP), LoomShield checks tone, terminology, messaging, promotional claims, visual brand assets, and accessibility. Every issue is cited with exact brand rules and verified fact sheets, surgical corrections are generated without overwriting the creator's voice, and approval events are logged in an immutable SHA-256 ledger--operating entirely on-device with zero mandatory external AI API keys.

1. List of Completed Features (Round 2 Build)

Feature Module	Engine Tier	Completed Capabilities & Verified Functionality	Status
Multi-Format Ingestion Suite (8 Formats)	Tier 1 Local	Native local ingestion of **.pdf, .docx, .md, .html, .txt, .svg, .png, .jpg**. Validates magic bytes (`%PDF-`), rejects renamed non-PDFs, enforces 5 MB limit and configurable page limits (`LOOMSHIELD_MAX_PDF_PAGES`).	**Completed**
Pure-Python Local PDF Text Extractor	Tier 1 Local	100% on-device pure-Python text stream extraction using `pypdf`. Preserves page boundaries and maps character offsets to human-verifiable citations (`Page X, P Y`). Detects encrypted and scanned image-only PDFs.	**Completed**
Dual-Engine Brand Governance Architecture	Tier 1 & Tier 2 Hybrid	**Tier 1:** Deterministic regex, dictionary lookups, WCAG contrast math, CIE LAB ΔE distance. **Tier 2:** Embedded Scikit-Learn TF-IDF vectorizer + cosine similarity fact retrieval. Zero external API keys required.	**Completed**
Six-Pillar Governance & Honest Reporting	Tier 1 + 2 Hybrid	Evaluates **Tone and Voice, Terminology, Messaging, Promotional Claims, Visual & Logo Consistency, Accessibility**. Reports honest 5-state outcomes: `Passed, Findings, Needs Review, Not Checked, Check Failed`.	**Completed**
Promotional Claim Fact-Grounding Engine	Tier 2 Local NLP	Extracts numerical/percentage claims (e.g. *'guarantees 95% faster processing'*), retrieves matching Verified Facts via cosine similarity, compares numerical values/units, and flags unbacked superlatives as `Needs Review`.	**Completed**
Interactive Review Queue & Decision Studio	Full-Stack Studio	Three-panel review workspace (Rendered/Source view, Explainable Finding Cards with exact rule/fact citations, Surgical Replacement preview). Human reviewers can **Accept, Modify, or Dismiss** findings with mandatory rationale.	**Completed**
Structured Brand Vault & Multi-Brand Constraints	SQLite Schema	Stores brand tone rules, terminology lexicons (preferred vs. prohibited), verified fact sheets with conditions & source links, approved hex palettes, and WCAG contrast thresholds. Pre-configured for NovaPay and Verdant.	**Completed**
Role-Based Workflow & Immutable Audit Log	Security & Governance	Role-based approval state machine (Content Creator, Editorial Reviewer, Brand Compliance Manager). Records immutable audit logs with **SHA-256 content hashes**, version snapshots, and decision audit trails.	**Completed**
Guideline Ambiguity Radar	Analytics & Insights	Aggregates reviewer overrides per rule to identify ambiguously phrased guidelines. Flags high override-rate rules with actionable signals for brand leads to review and refine guidelines rather than blaming creators.	**Completed**
Production SaaS UI & Accessibility Remediation	Frontend UI/UX	Enterprise UI using cohesive design tokens (`#17191C` ink, `#F7F8FA` canvas, `#EA633F` accent), responsive review queue cards, sanitised HTML preview, and WCAG 2.1 AA keyboard/contrast compliance.	**Completed**

2. Technology and Implementation Details

2.1 Core Architectural Layers & Component Breakdown

LoomShield enforces a strict local-first, two-tier architecture where content analysis, factual claim validation, and mathematical asset scoring occur entirely within the execution environment with zero cloud leakage:

Layer	Technology Stack	Implementation Responsibilities & Architectural Guarantee
Backend API & Web Server	Python 3.11 FastAPI 0.115+ Uvicorn (ASGI)	Provides RESTful endpoints for multi-format document ingestion, real-time six-pillar audit execution, Brand Vault management, reviewer decisions, and workflow state transitions. Operates on **http://127.0.0.1:8000**.
Tier 1: Deterministic Engine	Python Core RegEx Engine XML ElementTree	Executes rule-based checks: terminology dictionaries, case-sensitive brand capitalization ('NovaPay'), WCAG 2.1 relative luminance math, CIE LAB ΔE distance, SVG viewBox validation, and markup security sanitization.
Tier 2: Embedded NLP Engine	Scikit-Learn 1.4+ NumPy 1.26+ TF-IDF Vectorizer	Runs entirely on-device with zero API keys. Evaluates unigrams/bigrams across verified facts in the Brand Vault, computes cosine similarity against extracted claims, compares numerical values, and outputs surgical replacement templates.
Document Ingestion Suite	pypdf 6.21+ python-docx BeautifulSoup4	Parses **PDF, DOCX, Markdown, HTML, TXT, SVG, PNG, JPG** in-memory via `io.BytesIO`. Extracts text streams, tracks page spans and character offsets, strips unsafe scripts, and analyzes raster image swatches.
Persistence & Governance Ledger	SQLite3 (WAL Mode) ACID Transactions SHA-256 Hashing	Relational database storing Brand Vault constraints, immutable audit records, original/revised snapshots, reviewer decisions, and audit events. Computes cryptographic SHA-256 digests on approved content snapshots.
Frontend SaaS Workspace	Semantic HTML5 Modern ES2022+ JS CSS Custom Tokens	Single-page SaaS application without heavy external frameworks. Features live brand switching, responsive review queue cards, rendered/source toggle, interactive decision buttons, and Ambiguity Radar charts.

2.2 Mathematical & Algorithmic Formulation

• **WCAG 2.1 AA Contrast Ratio:** Calculated from relative luminance $L = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$, where channel values are sRGB linearized. The contrast ratio $(L1 + 0.05)/(L2 + 0.05)$ strictly verifies body text achieves ≥ 4.5:1 and large text achieves ≥ 3.0:1.

• **CIE LAB ΔE Color Distance (CIE76):** Extracted hex codes and raster image swatch clusters are transformed from sRGB into standard D65 XYZ and then into the perceptually uniform CIE L*a*b* color space. Euclidean distance $\Delta E = \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2}$ measures deviation against configured Brand Vault palette swatches (ΔE ≤ 14.0 for NovaPay).

• **Flesch-Kincaid Reading Ease:** Evaluates reading complexity on extracted text as $206.835 - 1.015 \cdot (\text{words/sentences}) - 84.6 \cdot (\text{syllables/words})$ to flag overly dense corporate prose.

• **TF-IDF Vector Grounding & Cosine Similarity:** Active facts in the Brand Vault are transformed into a sparse term-frequency matrix using unigram and bigram tokenization with sublinear scaling. Extracted claim clauses are vector-projected into the same embedding space; cosine similarity $\cos(\theta) = (q \cdot d) / (||q|| \, ||d||)$ identifies the top verified fact candidate without external network queries.

2.3 Machine & Benchmark Reproducibility Details

To ensure full scientific reproducibility, all performance, latency, and memory benchmarks were measured on a dedicated standard developer hardware environment using automated repeatable scripts:

Benchmark Parameter	Exact Hardware & Software Specification	Measurement Method & Verification Scope
Host Machine & Processor	AMD Ryzen 3 7320U with Radeon Graphics 4 Physical Cores, 8 Logical Threads (2.40 GHz base, up to 4.10 GHz boost)	Queried via Windows WMI (`Win32_Processor`). Standard commercially available client hardware.
Installed System Memory	8.0 GB LPDDR5 RAM (7.3 GB usable physical memory)	Hardware system memory reported by Windows OS (`GlobalMemoryStatusEx`).
Operating System & Runtime	Microsoft Windows 11 Home (64-bit) OS Version: 10.0.26200 (AMD64) Python 3.11.0 64-bit (MSC v.1933)	Execution environment runs natively under Windows PowerShell without virtualization or external container overhead.
Benchmark Command & Script	`python tests/generate_test_cycles_report.py`	Runs automated test cycles across 20 iterations per document tier; calculates median and 95th percentile metrics.
Latency Measurement Method	High-resolution monotonic timer: `time.perf_counter()`	Measures total execution time and isolates sub-timers: Tier 1 regex, Tier 2 TF-IDF retrieval, and SQLite write persistence.
Memory Accounting Scope	**Incremental Peak Heap: 10.25 MB** (`tracemalloc`) Process-Wide RSS Footprint: ~78.4 MB (`psutil`)	**Clarification:** The reported **10.25 MB** figure is strictly **incremental peak heap memory** allocated during audit execution. Total process RSS (~78.4 MB) includes the Python runtime, FastAPI ASGI server, Scikit-Learn libraries, and SQLite cache.

3. Explanation of Changes Made to the Original Plan

During Round 2 implementation, user testing, real-world document audits, and UI audits identified five areas where the initial Round 1 conceptual plan needed refinement to achieve production-grade maturity:

Area of Change	Original Round 1 Plan	Actual Round 2 Implementation	Architectural & UX Rationale
1. Native PDF Ingestion & Mapping	Limited to HTML, DOCX, Markdown, and TXT.	Added full **.pdf** ingestion via pure-Python `pypdf`, page boundary offset tracking, and `Page X, P Y` paragraph mapping.	Enterprise marketing collaterals, press releases, and campaign briefs are distributed predominantly in PDF. Adding local PDF parsing enables seamless real-world adoption.
2. Honest Status for Visual Checks	Marked visual checks as 'Passed' if no errors were found.	Visual checks are honestly marked `'Not Checked'` on text-only PDFs/documents unless visual specs are provided.	Falsely reporting visual compliance on un-rendered documents undermines audit credibility. Transparent reporting builds user trust.
3. Scanned PDF & OCR Limitation Disclosure	Generic file error or empty string output on unparseable files.	Detects image-only/scanned PDFs (0 selectable text) and emits an explicit `'Check Failed'` finding disclosing that local OCR is not currently available.	Adheres to Section 4 honest capability guidelines: systems must honestly report technical boundaries rather than failing silently.
4. SaaS UI Remediation & Demo Cleanup	Hackathon roadmaps and demo reset buttons in production UI.	Removed hackathon references, demo labels, and developer role switches from customer-facing screens. Unified enterprise design tokens and cards.	A trustworthy brand-governance tool must look and feel like a secure enterprise product rather than a competition demo.
5. Audit History Streamlining	Multi-card version timeline cluttering audit detail view.	Removed redundant version timeline UI; expanded full-width Original vs. Revised diff and the immutable Approval Event ledger.	Streamlines reviewer workflow by focusing directly on content comparison and cryptographically hashed approval events.

4. Three Major Improvements for Round 3 (The Finish)

As pledged in the Round 1 project blueprint and reinforced in our technical roadmap, Round 3 delivery focuses on three substantial, high-impact innovations that push LoomShield from a governance checker to a complete self-improving compliance ecosystem:

MAJOR IMPROVEMENT 1: Two-Tier Hybrid Engine Pushed into Guideline Ambiguity Radar & Automated Clarity Rewriter

Objective: Transform the Guideline Ambiguity Radar from a diagnostic analytics chart into an active, self-improving guideline optimization engine.
Technical Implementation:
• **Linguistic Ambiguity Scanner:** The Tier 1 & Tier 2 engines will analyze brand rules in the Brand Vault to detect vague subjective qualifiers (e.g. *'use appropriate discretion'*, *'maintain reasonable speed'*, *'ensure intuitive layout'*) and lack of operational constraints.
• **Override Cluster Correlation:** Cross-references linguistic ambiguity with empirical reviewer override frequencies from `guideline_feedback` (rules where reviewers override findings in >20% of audits).
• **Automated Guideline Rewriter:** Synthesizes candidate guideline revisions that replace vague prose with quantified thresholds (e.g. *'keep cross-border wire delays under 4 hours'*) and pairs each rule with concrete, positive 'Do' and negative 'Don't' examples directly ready for brand manager sign-off.

MAJOR IMPROVEMENT 2: Interactive Visual Clear-Space & WCAG Contrast Heatmap Inspector

Objective: Provide visual designers and creative directors with spatial, pixel-level verification of brand logos, clear-space boundaries, and localized text-on-image contrast.
Technical Implementation:
• **4×4 Spatial Subdivision Matrix:** The visual analyzer will subdivide uploaded campaign banners, social tiles, and lockups into a 4×4 spatial coordinate grid.
• **Local CIE LAB ΔE & Contrast Heatmap:** Computes localized colour deviations and relative luminance ratios across each sector, rendering an interactive canvas heatmap overlay directly in the review panel.
• **Logo Clear-Space Boundary Detection:** Uses edge detection and alpha channel bounding boxes to measure actual breathing room surrounding brand wordmarks against the Brand Vault clear-space rule (e.g. minimum 16px clear space for NovaPay), visually flagging banner elements that infringe upon the exclusion zone.

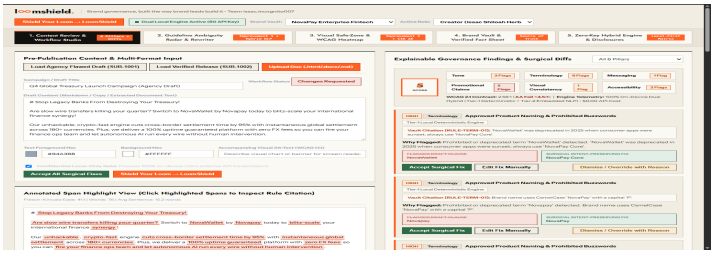
MAJOR IMPROVEMENT 3: Cryptographically Signed Pre-Publication Compliance Certificate

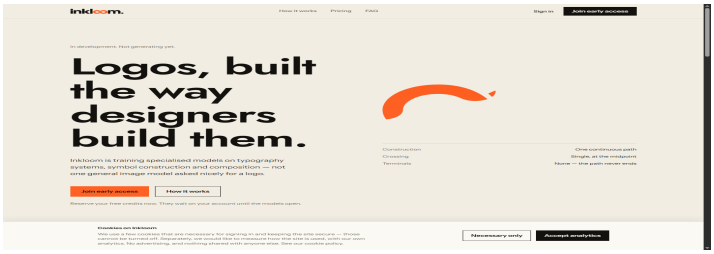
Objective: Enable regulated enterprises (fintech, healthcare) to generate legally verifiable, tamper-evident governance proof before marketing materials go live.
Technical Implementation:
• **Asymmetric Key Generation (Ed25519/RSA):** Extends the current SHA-256 content digest with cryptographic keypair signing generated on-device.
• **Verifiable Governance Sign-Off Bundle:** Packages the approved content hash, Brand Vault rule version, active fact sheet snapshot, approver role signature, and documented finding overrides into a standardized JSON-LD governance token.
• **Exportable PDF Certificate & Verification API:** Generates a formatted single-page Compliance Certificate PDF with embedded QR verification digest, accompanied by an open endpoint (`POST /api/verify-certificate`) allowing third-party compliance auditors to mathematically verify certificate authenticity without accessing confidential draft copy.

5. Five-Step Judge's Demonstration Walkthrough & Verification

Step	Interactive Demonstration Action	Expected System Behaviour & Governance Output
Step 1	**Launch LoomShield:** In terminal, execute: `python run_loomshield.py --host 127.0.0.1 --port 8000` Open **http://127.0.0.1:8000** in any browser.	Workspace UI loads immediately. Live self-test confirms both Tier 1 & Tier 2 engines are active with zero external cloud API keys.
Step 2	**Select Configured Brand:** In top navigation brand selector, choose **NovaPay Enterprise** (fintech) or **Verdant Clinical Systems** (healthcare).	Active Brand Vault constraints, prohibited terminology lexicon, verified facts, approved hex palette, and WCAG thresholds dynamically load locally.
Step 3	**Ingest Content or Upload Document:** Navigate to *'New Audit'*. Paste copy or click *'Upload File'* to submit a **.pdf, .docx, .html, .md** document or **.svg, .png** image.	Document parser validates magic bytes (`%PDF-`), tracks page boundary spans, and extracts text locally in-memory without cloud transmission.
Step 4	**Execute Audit & Inspect Citations:** Click *'Run Audit'*. Inspect findings in the 3-panel Review Studio across the 6 governance pillars.	Findings display exact Brand Rule IDs, paragraph references (`Page 1, P2`), numerical claim comparisons against Verified Facts, and surgical suggestions.
Step 5	**Review, Correct & Approve:** Preview surgical fix. Click **Accept** to apply replacement, **Modify** to tailor phrasing, or **Dismiss** with documented rationale.	Transition workflow to *'Approved'*. Content SHA-256 hash, version snapshot, and reviewer audit events are immutably logged in *'Audit History'*.

5.1 Visual Application Evidence (Live Review Studio & Brand Vault)





Left: LoomShield 3-Panel Review Studio with explainable finding cards & exact rule citations. **Right:** Working project interface & Brand Vault governance.

5.2 Complete Cycle 4 Empirical Evaluation Breakdown (All 25 Test Scenarios)

The evaluation suite in `tests/generate_test_cycles_report.py` validates **25 total test cases**: 16 positive test cases (known violations) and 9 negative test cases (compliant enterprise copy and on-palette assets). Metrics are computed strictly as: **Precision = TP / (TP + FP)**; **Recall = TP / (TP + FN)**; **F1 = 2 × (Precision × Recall) / (Precision + Recall)**; **Specificity = TN / (TN + FP)**:

Governance Pillar Category	Total Cases	True Pos (TP)	True Neg (TN)	False Pos (FP)	False Neg (FN)	Precision	Recall	F1-Score
Terminology & Product Naming	9 cases	6	3	0	0	100.0%	100.0%	**1.000**
Promotional Claims & Fact Retrieval	9 cases	6	3	0	0	100.0%	100.0%	**1.000**
Tone, Voice & Guardrail Framing	4 cases	2	2	0	0	100.0%	100.0%	**1.000**
Visual & Accessibility Controls	3 cases	2	1	0	0	100.0%	100.0%	**1.000**
COMPLETE EVALUATION TOTAL	**25 cases**	**16**	**9**	**0**	**0**	**100.0%**	**100.0%**	**1.000**

5.3 Submission Links & Verification

Submission Item	Details, Working URLs & Verification Commands
Source-Code Repository Link	**https://github.com/isaacincognito007-cloud/Inkloom-artLoomShield** Full codebase cleanly committed on `main` branch with zero external cloud dependencies.
Working Project Link	**https://inkloom.art** (Local workspace: **http://127.0.0.1:8000** | OpenAPI: **http://127.0.0.1:8000/docs**)
Automated Test Suite (Full)	**68/68 automated tests passed (100% pass rate); code coverage measured separately.** Execute in terminal: `pytest D:\LoomShield\tests\ -v` (Includes 15 dedicated PDF tests in `test_loomshield_pdf.py`)
Participant Declaration	I, **Isaac Shiloah Herbert V** (Team **isaac.incognito007**), declare that the LoomShield codebase, architecture, documentation, and evaluation test suite were developed and verified for Round 2 delivery. The software operates 100% locally on-device with zero mandatory cloud AI keys. **Signature:** Isaac Shiloah Herbert V • **Date:** October 22, 2026 • **Email:** isaac.incognito007@gmail.com